Guided Herbal Infusion Brewer

IDE 202.A

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Observation

People interested in herbal wellness often search online for recipes and health benefits. However, the large amount of information available about wellness and health, especially after Covid creates misinformation and information overload, making it difficult to identify reliable and healthy preparation methods.

Design Opportunity

There is an opportunity to design a product that guides users through the preparation process, helping them navigate complex information while supporting calm and mindful self-care rituals.

Design Problem

As a result, users who want to incorporate herbal infusions into their daily routines can feel uncertain, overwhelmed, and stressed when choosing how to prepare them safely and effectively.

Project Concept

A guided brewing machine that helps users prepare herbal infusions through controlled temperature and measured ingredients. Developed in response to misinformation and information overload around herbal wellness, the device provides curated healthy recipes and a simple interface to guide users toward mindful, stress-free preparation.

- Guided brewing
- Herbal wellness
- Information clarity
- Precision brewing
- Healthy Blends

Research Insight

Herbal infusions are commonly used for wellness purposes such as relaxation, digestion support, and immune health. However, many herbs contain active compounds that can affect the body when consumed in certain quantities.

Examples from research include:

- Chamomile may cause allergic reactions in some individuals
- Licorice root can raise blood pressure when consumed in large amounts
- Certain herbal infusions may interact with medications

Insight

Herbal teas are often perceived as harmless, but **they can have measurable physiological effects**, meaning preparation and dosage matter.

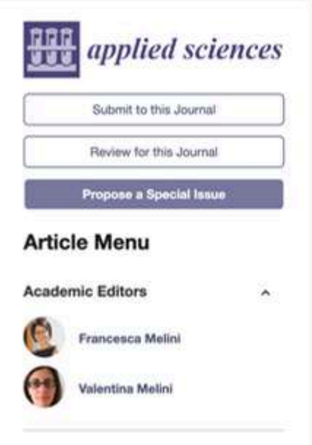
Research also shows that brewing conditions influence the compounds extracted from herbs, including:

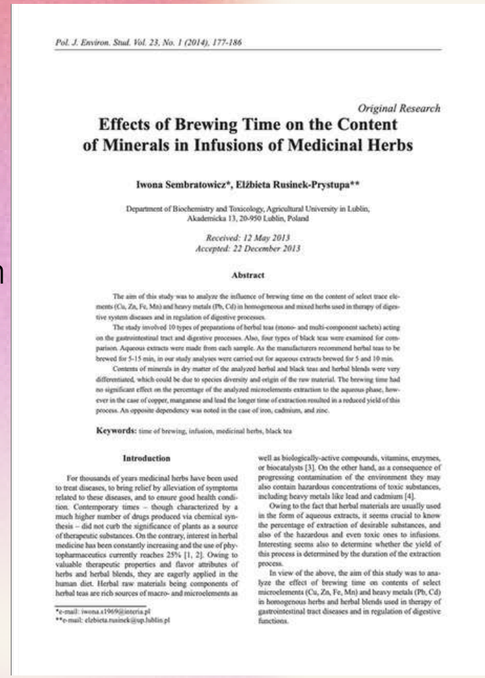
- water temperature
- herb-to-water ratio
- steeping time

Improper preparation may lead to weak, overly concentrated, or inconsistent infusions.

Insight

Herbal tea preparation benefits from controlled brewing conditions, which typical manual preparation methods do not provide.

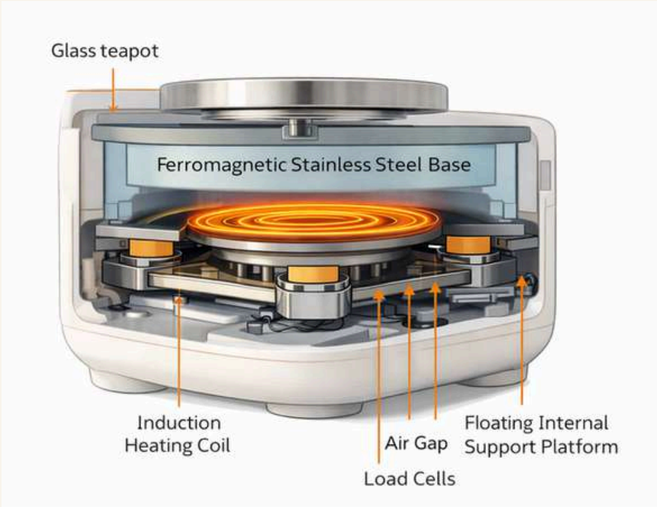




Technical Research: Brewing System Components

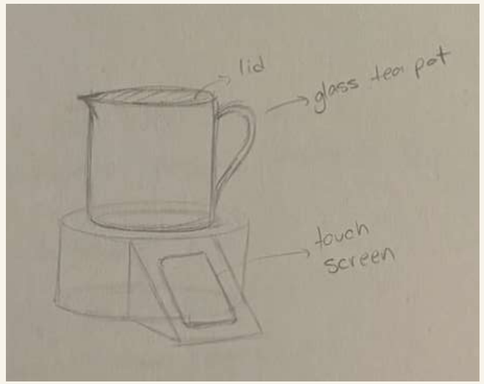
Before developing the form of the product, I first identified the technical systems required for the machine to function. To understand how such a device could operate, I examined existing appliances such as coffee machines and multi-functional kitchen devices like the Thermomix. Through this research, I identified the core components needed for the system: a weighing system (load cells) to measure ingredients, a heating system (initially explored as induction heating) to control water temperature, and a screen interface to guide the brewing process.

Because I was not fully familiar with all the technical systems involved, I used **AI** as a discussion tool at this stage a lot. We discussed possible solutions and system structures.

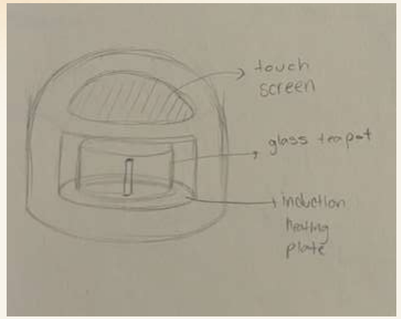


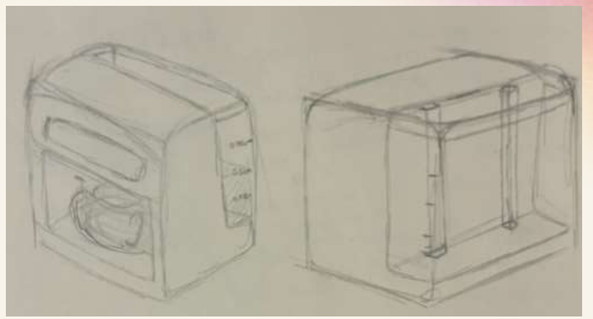
Initial Concept & Design Iteration

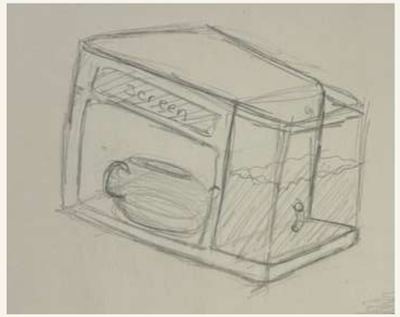
At first, I assumed that all systems could be solved within a very compact structure. However, this approach negatively affected the product's form, and later in the process I realized that several important elements were missing.

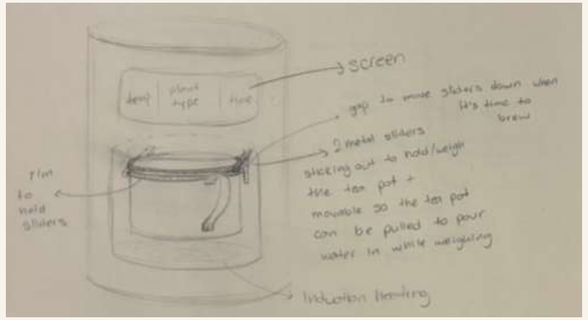


While doing further research I realized that to brew herbal tea you have to pour the already heated water over the loose leaves which changed my whole concept of the compact form and i started working on closed forms with a system that would heat the water before hand.



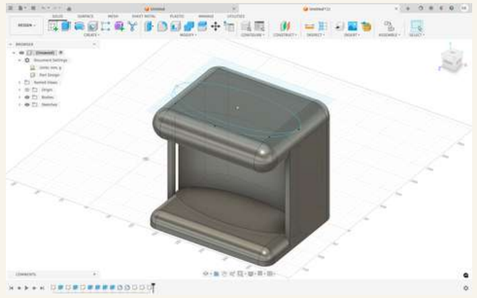


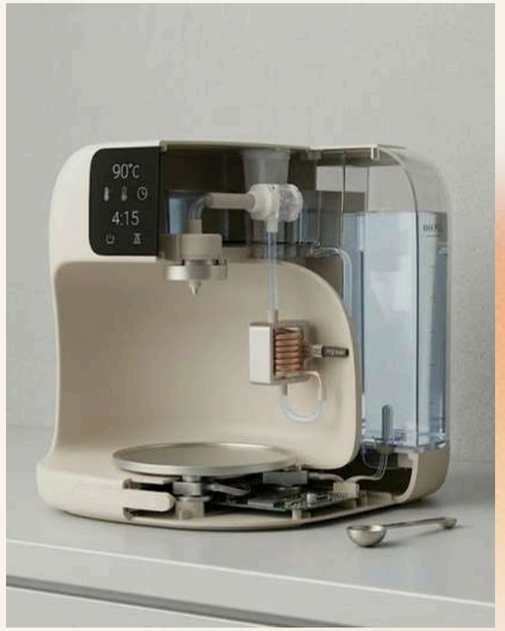




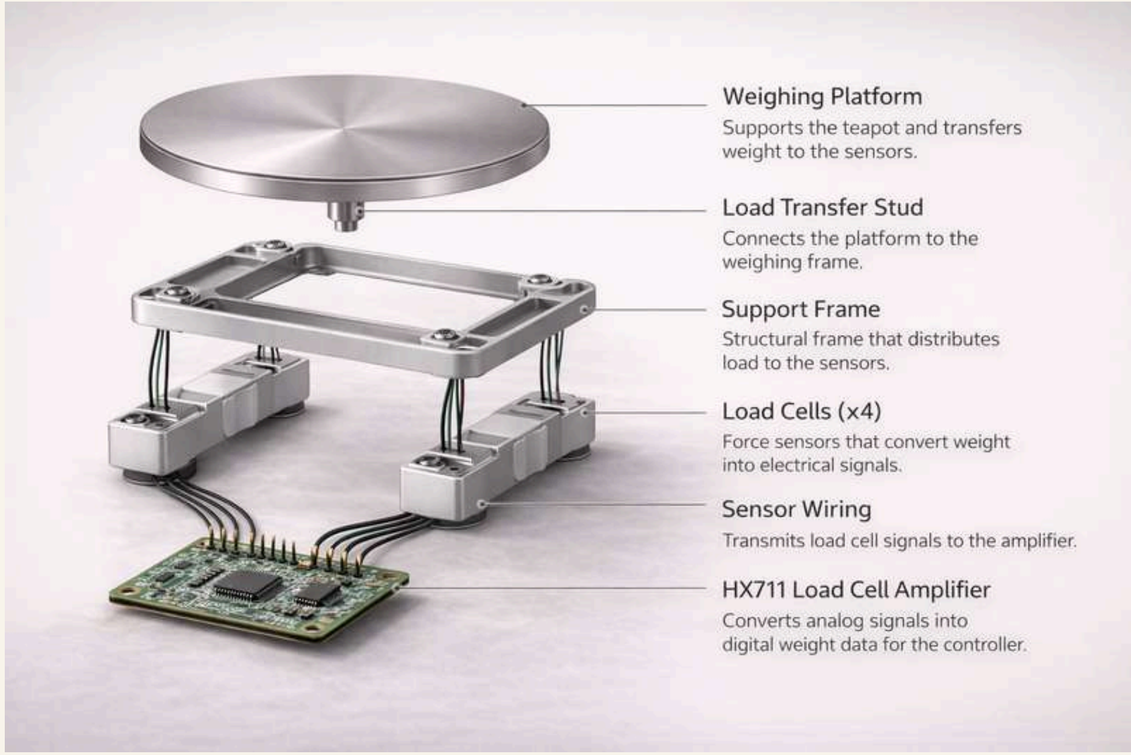
Form and System Exploration

At this stage I had a lot of discussions with Chat GPT trying to visualize thoughts and I struggled at this stage the most the AI sources wouldn't understand my sketches and struggled to make the adgustments I drew and asked for.





System Architecture



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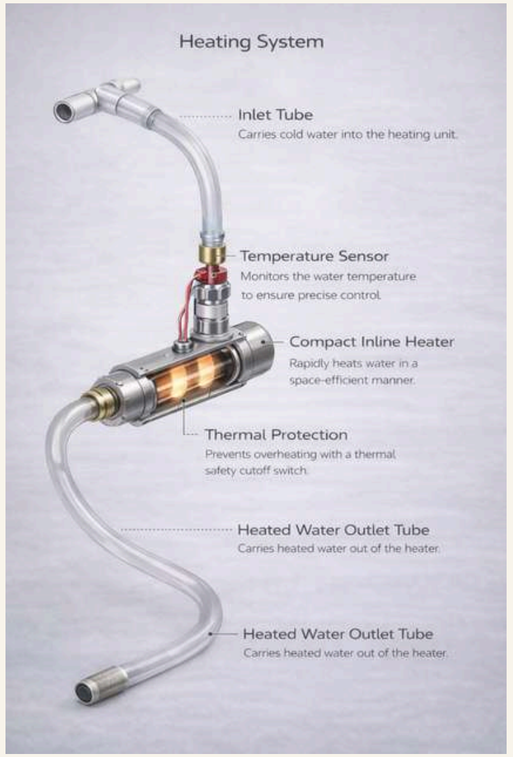

AI as a Design Partner

Prompt

After researching herbal brewing methods, I realized that pouring cold water over herbs and heating them together is not correct for infusions. The water should first be heated to the required temperature and then poured over the herbs. This means the system might need to change from heating the teapot directly to heating water separately and dispensing it afterward

AI Response Summary

The AI suggested restructuring the machine so that water is heated in a separate upper module and then dispensed into the teapot once the herbs are placed.

My Reflection

This interaction significantly changed the architecture of the product. The machine shifted from a heated teapot concept to a controlled hot-water infusion system, which better reflects how herbal infusions are actually prepared. But the systems AI suggested were not the most productive systems, so I took the suggestion and made adgustments and changes to make it more feasible.

Re-framer

Prompt

I am trying to define my user group. I am thinking about women aged 23–35 who are interested in wellness rituals they see on social media, but I am struggling to express the problem clearly. What I am trying to say is that many people lack reliable scientific grounding for these practices. They often mislearn information, follow incorrect advice, or feel overwhelmed by the huge amount of conflicting information online, which makes it difficult to distinguish what is healthy or correct.

AI Response Summary

The AI suggested framing the user group as people interested in herbal wellness who are motivated by social media rituals but lack reliable guidance. It emphasized that the core problem is not simply interest in herbal tea, but information overload and misinformation, which makes it difficult for users to confidently prepare healthy infusions.

My Reflection

This interaction helped me clarify the real design problem behind the project. Instead of focusing only on making a tea brewing machine, I began framing the product as a guided brewing system that helps users navigate misinformation and simplify herbal preparation.

Clarifier

Final Product

Menu Interface

Brewing Interface

App Interface:

Select Herbs

Recipe

Recipe Options